

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution				
Obtains one correct missing expected value e.g. 0 and car = $\frac{28 \times 35}{80}$	AO1.1a	M1		0	1	2	3+
			C	12.25	10.0625	8.3125	4.375
			MB	15.75	12.9375	10.6875	5.625
Obtains all expected values Must not be rounded CAO Condone 12.937 for 12.9375	AO1.1b	A1					

Marking Instructions	AO	Marks	Typical Solution												
States both hypotheses using correct language Accept equivalent wording	AO2.5	B1	H_0 : There is no association between number of claims and type of insurance policy H_1 : There is an association between number of claims and type of insurance policy												
Combines expected and observed values for 2 and 3 or more correctly Follow through their answers to part (a)	AO1.1b	B1F	<table><tr><td></td><td colspan="2">2 or more</td></tr><tr><td></td><td>Observed</td><td>Expected</td></tr><tr><td>C</td><td>16</td><td>12.6875</td></tr><tr><td>MB</td><td>13</td><td>16.3125</td></tr></table>		2 or more			Observed	Expected	C	16	12.6875	MB	13	16.3125
	2 or more														
	Observed	Expected													
C	16	12.6875													
MB	13	16.3125													
Calculates χ^2 -test statistic Can be awarded even if 2 and 3 or more are not combined	AO1.1a	M1	$\sum \frac{(O - E)^2}{E} =$ $\frac{(9 - 12.25)^2}{12.25} + \frac{(19 - 15.75)^2}{15.75} +$ $\frac{(10 - 10.0625)^2}{10.0625} + \frac{(13 - 12.9375)^2}{12.9375} +$ $+ \frac{(16 - 12.6875)^2}{12.6875} + \frac{(13 - 16.3125)^2}{16.3125}$ $= 3.07$												
Calculates χ^2 -test statistic correctly Condone given to 1 s.f. CAO	AO1.1b	A1													
States critical value (or p -value, follow through their χ^2 value) Condone χ^2 cv for 3 df = 6.251 if 2 and 3 or more are not combined	AO1.1b	B1F	χ^2 cv for 2 df = 4.605 ($p = 0.025$) $3.07 < 4.605$ ($0.025 > 0.01$) Accept H_0												
Evaluates χ^2 -test statistic by comparing the cv with their ts (or their p value with 0.1)	AO3.5a	R1	No significant evidence to suggest that there is an association between number of claims and type of insurance policy												
Infers H_0 not rejected (OE)	AO2.2b	E1													

Concludes in context (not definite) Accept equivalent wording	AO3.2a	E1	
Total 10 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution																				
(a)	Obtains one missing expected value eg Low and Breed A = $\frac{21 \times 27}{105}$	AO1.1a	M1	<table><tr><th></th><th>Low</th><th>Med</th><th>High</th></tr><tr><td>A</td><td>5.4</td><td>9.6</td><td>6</td></tr><tr><td>B</td><td>5.14</td><td>9.14</td><td>5.71</td></tr><tr><td>C</td><td>8.23</td><td>14.63</td><td>9.14</td></tr><tr><td>D</td><td>8.23</td><td>14.63</td><td>9.14</td></tr></table>		Low	Med	High	A	5.4	9.6	6	B	5.14	9.14	5.71	C	8.23	14.63	9.14	D	8.23	14.63	9.14
		Low	Med		High																			
A	5.4	9.6	6																					
B	5.14	9.14	5.71																					
C	8.23	14.63	9.14																					
D	8.23	14.63	9.14																					
	Obtains all expected values. AO if integers	AO1.1b	A1																					
(b) (i)	States both hypotheses using correct language	AO2.5	B1	H_0 Milk yield is independent of breed H_1 Milk yield is not independent of breed 1 tail 1% $ts = \sum \frac{(O - E)^2}{E} = 19.4$ χ^2 cv for 6 df = 16.81 ($p = 0.00275$) 219.4 > 16.81 (0.00275 < 0.01) Reject H_0 and conclude that there is evidence to suggest that milk yield is not independent of breed																				
	States critical value (or p -value)	AO1.1b	B1																					
	Evaluates the χ^2 test statistic by comparing the cv with the ts (or p -value with 0.01)	AO3.5a	R1																					
	Infers H_0 rejected	AO2.2b	E1																					
	Concludes correctly in context (conclusion should not be definite)	AO3.2a	E1																					
(ii)	Considers $\frac{(O - E)^2}{E}$ or $(O - E)$ to identify largest sources of association [Do not allow mark if no reference to why source selected]	AO2.4	E1	Largest sources of association Breed A/High and Breed B/ Low. Far more than expected Breed A cows observed to have high milk yield or Far more than expected																				
	Interprets main source(s) of association in context	AO3.2a	E1dep																					

			Breed B cows observed to have low milk yield
Total 9 marks			

Q3.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.2	B1	8
Total 1 mark			

Q4.

Marking Instructions	AO	Marks	Typical Solution																
States both hypotheses using correct notation	AO2.5	B1	H ₀ : Language choice is independent of sex																
Constructs 'correct' contingency table with frequencies shown (PI)	AO3.1b	M1	H ₁ : Language choice is not independent of sex <table><tr><td></td><td>German</td><td>Mandarin</td><td></td></tr><tr><td>M</td><td>7</td><td>12</td><td>19</td></tr><tr><td>F</td><td>23</td><td>8</td><td>31</td></tr><tr><td></td><td>30</td><td>20</td><td>50</td></tr></table>		German	Mandarin		M	7	12	19	F	23	8	31		30	20	50
	German	Mandarin																	
M	7	12	19																
F	23	8	31																
	30	20	50																
Finds expected frequencies (PI)	AO3.4	B1	<table><tr><td></td><td>German</td><td>Mandarin</td></tr><tr><td>M</td><td>11.4</td><td>7.6</td></tr><tr><td>F</td><td>18.6</td><td>12.4</td></tr></table>		German	Mandarin	M	11.4	7.6	F	18.6	12.4							
	German	Mandarin																	
M	11.4	7.6																	
F	18.6	12.4																	
Evaluates $\chi^2 = \sum \frac{(O - E)^2}{E}$ Allow no Yates correction used FT their observed and expected frequencies	AO1.1a	M1F	$\chi^2 = \sum \frac{(O - E - 0.5)^2}{E}$ $= \frac{(3.9)^2}{11.4} + \frac{(3.9)^2}{7.6} + \frac{(3.9)^2}{18.6} + \frac{(3.9)^2}{12.4}$																
Obtains χ^2 test statistic or quotes <i>p</i> -value	AO1.1b	A1	$= 1.33 + 2.00 + 0.82 + 1.23$ ts $\chi^2 = 5.38$ 0.0204																
Evaluates χ^2 model by comparing test statistic and correct critical value Or by comparing <i>p</i> -value with 0.05	AO3.5a	M1	$p = 0.0204$ 1 tail 5% df = 1 3.84 cv = 3.84 or $p < 0.05$ ts >																
Infers that H ₀ should be rejected	AO2.2b	E1	Reject H ₀																

Completes test correctly and gives conclusion in context. (conclusion should not be definite)	AO3.2a	E1	Significant evidence to suggest that language choice is not independent of sex
Total 8 marks			

Q5.

(a)

Expected values				
	F	T	S	D
<3	8.736	34.944	24.192	16.128
>3	4.264	17.056	11.808	7.872

Any two correct to 2 d.p.

M1

All correct, here or below, to 2 d.p.

A1

*Must be **expected** value, applied to this case, not just general statement.*

E1

One expected value for Flat < 5
So combine first two columns to give

Expected values			
	F+T	S	D
<3	43.68	24.192	16.128
>3	21.32	11.808	7.872

For combining first two E columns, at least 1 correct.

B1

H_0 : No association between property type and time to sell.

H_0 seen somewhere in solution

H_1 : Association between property type and time to sell.



If "independent" used then must be correct way round

B1

O_i	E_i	$(O_i - E_i)^2 / E_i$
38	43.68	0.7386
27	21.32	1.5132
28	24.192	0.5994
8	11.808	1.2281
18	16.128	0.2173
6	7.872	0.4452
	χ^2	4.7418

For attempt at $(O_i - E_i)^2 / E_i$

M1

$$4.7 < \chi^2 < 4.8$$

A1

2 degrees of freedom

PI by 4.605, 5.991, 7.378, 9.210 or 10.597 seen

B1

C.V. of χ^2 for 2 d.f. = 4.605

AWFW 4.60 to 4.61

B1

4.74 > 4.605 so reject H_0
significant evidence of an association
between property type and time to sell.

Context conclusion.

Dep. on B1 for H_0 , A1 for χ^2 and
B1 for c.v.

A1

10

- (b) (i) More in total than any other type so likely to have biggest effect
Or similar referring to large number

E1

- (ii) Far away from expected values
Or opposite pattern to other three

E1

2

[12]

If Flats and Detached combined:

Expected values			
	F+D	T	S
<3	24.864	34.944	24.192
>3	12.136	17.056	11.808

For combined F and D

B1

O_i	E_i	$(O_i - E_i)^2 / E_i$
22	24.864	0.3299
34	34.944	0.0255
28	24.192	0.5994
15	12.136	0.6759
18	17.056	0.0522
8	11.808	1.2281
	χ^2	2.9110

For attempt at $(O_i - E_i)^2 / E_i$

M1

A0

Accept H^0

A0

Max of 8 marks

2

Q6.

(a)

O_i	E_i	$(O_i - E_i - 0.5)^2 / E_i$
30	35.2	0.6276
14	8.8	2.5102
130	124.8	0.1770
26	31.2	0.7080
	χ^2	4.0228

E attempted (at least two correct to 1 d.p.)

M1

Yates' correction attempted; at least one correct value in final column

M1

χ^2 attempted

M1

AWFW 4.02 to 4.03

A1

H_0 : No association between method of receiving information and outcome
At least one correct

H_1 : Association between method of receiving information and outcome
If "independent" used, it must be the right way round

B1

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 CV of χ^2 for 1 df = 3.84(1)

B1

4.02 > 3.841 so reject H_0
 There is significant evidence of an association between method of receiving information and outcome
Dep on A1 and B1 for CV

A1

Applications higher than expected for telephone calls, so council's belief seems to be true
Dep on previous A1
Context conclusion about council's belief, referring to higher than expected for telephone

Adep1

Alternative if Yates' not used

O_i	E_i	$(O_i - E_i)^2 / E_i$
30	35.2	0.7682
14	8.8	3.0727
130	124.8	0.2167
26	31.2	0.8667
	χ^2	4.9243

Loses M1 for Yates' and A1 for final χ^2 value but can score all the other 6 marks

Final 2 A1 marks dep on 4.92 to 4.93 and B1 for CV

- (b) Type I error was made because

E1

H_0 has been rejected (when it was true)

Dep on previous E1

Edep

SC If ' H_0 accepted' when
their χ^2 less than their CV

No error was made because

H_0 has been accepted (when it was true)

(E1)

Dep on previous (E1)

(Edep1)

2

[10]

Q7.

- (a) (i) $E_i: \frac{mp}{N}; \frac{mq}{N}; \frac{np}{N}; \frac{nq}{N}$

B1 any one correct

B2 all correct (simplified)

B2,1

2

- (ii)

$$\begin{aligned}\sum_i E_i &= \frac{mp + mq + np + nq}{N} \\ &= \frac{m(p+q)}{N} + \frac{n(p+q)}{N} \quad (\text{oe}) \\ \sum_i E_i &= \frac{mp + mq + np + nq}{N} \\ &= \frac{m(p+q) + n(p+q)}{N}\end{aligned}$$

(or use of unsimplified forms)

M1

$$\frac{mN}{N} + \frac{nN}{N}$$

$$= m + n$$

$$= \frac{(p+q)(m+n)}{N} = \frac{N \times N}{N} = N$$

Mdep1

$$= N \text{ (since } p + q = m + n = N \text{)} \\ \text{(AG)}$$

Adep1

An example of unsimplified values derived from $a = \frac{mp}{N}$:

$$\Rightarrow b = m - \frac{mp}{N}; c = p - \frac{mp}{N};$$

$$d = n - \frac{mp}{n} \quad (\text{oe})$$

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3

(b) H_0 : No association between Andy's results and wind conditions

B1

E_i :

17.82	15.18	33
9.18	7.82	17
27	23	50

Attempt E's

M1

$$\Rightarrow |0_i - E_i| - 0.5 = 2.32$$

Yates' correction attempted

M1

$$\chi^2 = 0.3020 + 0.3546 + 0.5863 + 0.6883$$

Final column attempted

M1

$$= 1.93$$

awrt

A1

$$\chi^2_{10\%}(1) = 2.706$$

correct value of χ^2 only (allow 2.71)

B1

$\Rightarrow$ Accept H_0

dep (B1 for H_0)

Adep1

No association (between Andy's results and wind conditions)

Appropriate conclusion

dep(B1 for H_0 ; M1final column; $\chi^2_{10\%} = 2.706$.)

Edep1

8

[13]

Q8.

(a) H_0 : No association between A level grade and class of degree

H_1 : Association between A level grade and class of degree

At least H_0 correct

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B1

O_i	E_i
20	11.6
9	17.4
36	36.4
55	54.6
22	28
48	42
2	4
8	6
200	200

For E_i 's attempted

M1

Combine Class 2(ii) and 3

For combining attempted

M1

20	11.6	8.4	6.0827
9	17.4	-8.4	4.0552
36	36.4	-0.4	0.0044
55	54.6	0.4	0.0029
24	32	-8	2.0
56	48	8	1.3333
200	200	0	13.47

For final column attempted

M1

(awrt 13.5)

A1

$\nu = 2$

[$\nu = 3$ with $\chi^2 = 11.345$ (B0B1ft)]

B1

$\chi^2_{1\%}(2) = 9.210$

B1

Reject H_0

Dep on B1 M1M1M1 B1B1, not A1

A1

Fiona's belief justified

Dep on B1 M1M1M1 B1B1, not A1

E1

9

- (b) **Fewer than expected gained a Class 1**
degree having gained grade B in A-level Mathematics.

E1

More than expected gained a Class 2(ii)
degree having gained grade B in A-level Mathematics.

E1

correct comments (see below)

1	2(i)	2(ii)	3	comb
9	55	48	8	56
17.6	54.6	42	6	48
A	B	C	D	E

A: fewer than expected

B: as expected

C: more than expected

D: more or similar than expected

E: more than expected

Q9.

(a)

	C	L	LD	OP	Tot
M	156	144	120	60	480
F	216	135	108	81	540
Tot	372	279	228	141	1020

B1

For each correct row

B1

2

(b) H_0 : No association between gender and the way students vote

H_1 : Association between gender and the way students vote

For at least H_0 correct

B1

O_i	E_i	$(O_i - E_i)^2 / E_i$
156	175.06	2.075
216	196.94	1.844
144	131.29	1.230
135	147.71	1.093
120	107.29	1.505
108	120.71	1.337
60	66.35	0.608
81	74.65	0.541
		$X^2 = 10.233$

Attempt at E_i

M1

Attempt at $(O_i - E_i)^2 / E_i$

M1

Attempt at
$$\sum \left(\frac{(O_i - E_i)^2}{E_i} \right)$$

M1

awfw 10.2 to 10.3

(A1 dependent on all 3 method marks)

A1

$$v = 3 \Rightarrow \chi^2_{\text{crit}} = 11.345$$

B1

ft on their v value

B1 ft

$$X^2 < \chi^2_{\text{crit}} \therefore \text{accept } H_0$$

A1

Accept claim at 1% level.
Evidence to suggest that the way students
vote is independent of gender.

E1

9

[11]

Q10.

- (a) H_0 : no association (between type of school and performance of 16 year olds in their GCSEs)

*H_0 : type of school and performance of 16 year olds
in their GCSEs independent*

B1

1

(b)
$$\frac{(O - E)^2}{E}$$

0.195819311
0.482160711
0.003569447
1.080536181

0.062507172
1.269422099
0.785491128
0.183802623

0.541856652
0.044011976
3.274102564
4.096492891

$$\frac{(O - E)^2}{E}$$

'Attempt at
(≥ 4 correct values seen to 2dp)

M1

$$X^2 = \sum \frac{(O - E)^2}{E}$$

Attempt to add ≥ 8 terms

m1

$$= 12.01977275$$

$$= 12.0 \text{ (1dp)}$$

$$\text{Allow } 11.9 \leq X^2 \leq 12.1 \Rightarrow M1m1$$

CAO

A1

3

(c) $v = 6 \Rightarrow \chi^2_{1\%} = 16.8(12)$

$$v = 6 \text{ can be implied by } \chi^2_{1\%} = 16.8(12)$$

B1,B1

No (significant evidence to suggest an) **association between**
(type of) **school and (GCSE) performance** (of 16 year olds)

Insufficient/no evidence to support Emily's belief.

School and performance are independent.

*Correct conclusion **in context***

*Dep on B1M1m1B1B1 given in (a), (b), (c) **and***

$$11.9 \leq X^2 \leq 12.1$$

Adep1

3

- (d) More than expected gained at least / more than 5 GCSEs
Fewer than expected gained at least / more than 1 GCSE
but less than 5 GCSEs
Fewer than expected gained no GCSEs

*Since conclusion of **no association** between school
and GCSE performance, it may be misleading to look at*

individual differences in any great detail

Any one of these 4 comments seen

B1

1

(e) $\chi^2_{10\%} = 10.6(45)$

Correct value of χ^2 only

B1

Reject H_0 at 10% level of significance.

(Evidence to suggest) an **association between** (type of)
school and (GCSE) performance

Evidence to support Emily's (Joanne's) belief.

(Type of) school + (GCSE) performance dependent.

*Dep on B1M1m1 **and** $11.9 \leq X^2 \leq 12.1$ **and** B1 in (e)*

Bdep1

2

Q11.

(a) H_0 : no association between age and first time performance in driving test

H_1 : association between age and first time performance in driving test

B1

	Pass		Fail		
Age	O	E	O	E	Total
17-18	28	19.2	20	28.8	48
19-30	2	6.4	14	9.6	16
31-39	12	18.0	33	27.0	45
40-60	6	4.4	5	6.6	11
Total	48	48	72	72	120

E's attempted

Correctly

M1

A1

O	E	$\frac{(O-E)^2}{E}$
28	19.20	4.0333
2	6.40	3.0250
18	22.40	0.8643
20	28.80	2.6889
14	9.6	2.0167
38	33.6	0.5762
		13.20

Attempt at combining

Correctly

M1

A1

Final column attempted

m1

For X^2 correct

A1

$$v = 2 \quad \Rightarrow \quad \chi^2(2) = 9.210$$

(on $v = 2$ or $v = 3$ only)

B1ft

Reject H_0
Evidence to support Julie's belief
at 1% level of significance.

E1ft

9

- (b) More students than expected in the age group 17-18 pass their test first time.

Fewer than expected fail

E1

1

[10]

Q12.

O_i	E_i	$(O_i - E_i - 0.5)$ (α)	α^2 / E_i
24	28	3.5	0.4375
56	52	3.5	0.2356
11	7	3.5	1.7500
9	13	3.5	0.9423
			3.3654

E attempted

M1

Yates' correction attempted

M1

χ^2 attempted

M1

A1

H_0 : No association between drug and prevention of sickness

H_1 : Association between drug and prevention of sickness

(at least H_0 stated correctly)

B1

$$\chi^2_{5\%} = 3.841$$

CAO

B1

Accept H_0

A1ft

No evidence at the 5% level of significance to support the claim that the drug is effective against sickness.

E1ft

8

[8]

Q13.

H_0 : No association between choice of subject and gender

H_1 : Association between choice of subject and gender

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B1

	Bul	Cl	Fin	Pol	Total
Male	7	31	25	40	103
Female	2	24	22	19	67
Total	9	55	47	59	170

Totals

B1

O_i	E_i
7	5.45
2	3.55

31	33.32
24	21.68
25	28.48
22	18.52
40	35.75
19	23.25
170	170

E's attempted (correctly)

M1A1

One of the E_i 's $< 5 \therefore$ combine cells

Attempt at combining (correctly)

M1A1

O_i	E_i	$\alpha = (O_i - E_i)$	α^2 / E_i
47	41.20	5.8	0.8165
21	26.80	-5.8	1.2552
31	33.32	-2.32	0.1615
24	21.68	2.32	0.2483
25	28.48	-3.48	0.4252
22	18.52	3.48	0.6539

Final column

m1

Test statistic: $X^2 = 3.56$

(AWFW 3.55 to 3.57)

A1

Critical value: = 4.605

ft on their v

B1F

Accept H_0

A1F

Insufficient evidence to suggest that the choice of subject is associated with gender.

E1

11

[11]

Q14.

H_0 : no association between age and attitude to school reorganisation

H_1 : association between age and attitude to school reorganisation

B1

Age	Against	
	O_i	E_i
16 – 17	9	$6 \frac{17}{65}$
18 – 21	17	$15 \frac{24}{65}$
22 – 49	115	$116 \frac{9}{13}$
50 – 65	41	$42 \frac{9}{13}$
> 65	3	$3 \frac{64}{65}$
Total	185	185

Age	Not Against	
	O_i	E_i
16 – 17	2	$4 \frac{48}{65}$
18 – 21	10	$11 \frac{41}{65}$
22 – 49	90	$88 \frac{4}{13}$
50 – 65	34	$32 \frac{4}{13}$
> 65	4	$3 \frac{1}{65}$
Total	140	140

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E's attempted correctly (at least 6 E's)

E_i
6.262
15.369
116.692
42.692
3.985

E_i
4.738
11.631
88.308
32.308
3.015

M1A1

Row totals: **11,27** 205, **75.7** (325)

Column totals: 185, 140 (325)

Totals correct

B1

E_i 's < 5

∴ combine cells 16 – 17 and 18 – 21 **also**

Attempt at combining rows

M1

50 – 65 and 'over 65' to give:

Correctly

A1

O_i	E_i	$a = O_i - E_i$	$\frac{a^2}{E_i}$
26	21.63	4.369	0.8825
115	116.69	-1.692	0.0245
44	46.68	-2.677	0.1535
12	16.37	-4.369	1.1662
90	88.31	1.692	0.0324
38	35.32	2.677	0.2029
325	325		2.462

Final column attempted (dep M1)

m1

$X^2 = 2.462$

2.4 to 2.5

A1

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$v = 2$

B1

$\chi^2_{v-2}(0.95) = 5.991$

On their v

B1ft

Accept H_0

A1ft

No real evidence at 5% level of significance to suggest any association between age and attitude to school reorganisation.

(context)

E1ft

Q15.

(a) H_0 : no association between education and salary

B1

O	E	$\alpha = O - E - 0.5$	$\frac{\alpha^2}{E}$
78	70.2	7.3	0.7591
57	64.8		0.8224
52	59.8		0.8911
63	55.2		0.9654
			3.4380

E attempted, correctly

M1

Yates' correction attempted

A1M1

$\frac{\alpha^2}{E}$ attempted (final col)

M1

AWRT 3.4

A1

$$\chi^2(10\%) = 2.706$$

B1

Reject H_0 at 10% level

A1ft

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Evidence to suggest an association between salary and having a university education.

E1ft

9

(b) Rejecting H_0 when H_0 correct

E1

Stating that there is an association between salary and education when there is not.

E1

2

[11]

Q16.

(a)

O_i	E_i	$ O_i - E_i - 0.5$	$\frac{7.5^2}{E_i}$
52	44	7.5	1.2784
58	66	7.5	0.8523
28	36	7.5	1.5625
62	54	7.5	1.0417
			4.7349

E attempted

M1

Yates' correction attempted

M1

χ^2 attempted

M1

AWFW 4.73 to 4.74

A1

H_0 : No association between incidence
of asthma and volume of traffic

H_1 : Association

at least H_0 stated correctly

B1

$\nu = 1$

$\chi^2_{\text{crit}} = 3.841 < 4.7349$

critical value

B1

Reject H_0 at 5% level

A1ft

Evidence to suggest an association
between the incidence of asthma in
children and the volume of traffic where they live

E1ft

8

(b) More than expected had asthma

dep on statement of association

E1

1

[9]

Q17.

- (a) H_0 : No association between the performances at KS3 and GCE

B1

O_i	E_i	$O_i - E_i$	X^2
60	63.55	-3.55	0.1983
55	44.64	10.36	2.4043
40	46.81	-6.81	0.9907
55	51.25	3.75	0.2744
32	36.00	-4.00	0.4444
38	37.75	0.25	0.0017
47	46.33	0.67	0.0097
31	32.54	-1.54	0.0733
35	34.13	0.87	0.0222
43	43.87	-0.87	0.0173
26	30.82	-4.82	0.7527
38	32.31	5.69	1.0005

E_i M1

$O_i - E_i$ M1

$(O_i - E_i)^2/E_i$

Σ M1

$$X^2 = 6.1897$$

AWFW 6.05 – 6.35

A1

$$v = 3 \times 2 = 6 \Rightarrow \chi^2_{90\%} = 10.645$$

on their v

B1B1ft

Do not reject H_0

No evidence to suggest an association between KS3 results and GCE grades at 10% level of significance.

E1ft

9

- (b) More of the students achieving level 7 at KS3 gain grade A's at GCE than expected.

E1

1

[10]

Q18.

- H_0 : condition independent of treatment
 H_1 : condition dependent upon treatment

B1

Totals: 66, 84, 75, 75

B1

O	E	$ O - E - 0.5$	$\frac{(O - E - 0.5)^2}{E}$
20	33	12.5	4.7348
55	42		3.7202
46	33		4.7348
29	42		3.7202

for E_i attempted, correctly

M1A1

for use of Yates' correction

M1

final column

M1

$$\chi^2 = 16.91$$

allow 16.9

If no Yates' correction:

possible M1A1M0M1A0

If 0.5 incorrectly used:

possible M1A1M1M1A0

A1

$$\chi^2_{0.05}(1) = 3.841 < 16.91$$

for χ^2 on their v

B1ft

Reject H_0

iff H_0 stated correctly

dependent on third M1

A1ft

Evidence to suggest that the condition of the patients may be dependent upon the treatment that they received

E1ft

10

[10]

Q19.

(a) (i)

	A	B	Total
--	---	---	-------

22-34	21	32	53
35-39	72	36	108
40-59	27	12	39
Total	120	80	200

for A values

B1

for B values

B1

2

(ii) H_0 : no association between area and age profile.

H_1 : association between area and age profile.

B1

At least H_0 .

O_i	E_i	
21	31.8	3.6679
72	64.8	0.8000
27	23.4	0.5538
32	21.2	5.5019
36	43.2	1.2000
12	15.6	0.8308
$\sum O_i = 200$	$\sum E_i = 200$	$\chi^2 = 12.554$

attempt row & column totals

M1

attempt at E_i

M1

attempt at $\frac{(O_i - E_i)^2}{E_i}$

M1

attempt at χ^2

M1

(AWFW 12.5 to 12.6)
provided correct method used

A1

$$v = (3 - 1)(2 - 1) = 2$$

B1

$$\chi^2_{1\%}(2) = 9.210 < 12.554$$

on their v and χ^2

B1ft

Reject H_0

The evidence suggests that the area within which a school is situated seems to have an effect on the age-profile of the staff employed.

on χ^2 and calculated value dep on H_0 correct, if stated.

E1ft

9

- (b) There seems to be fewer staff employed in 22 - 34 age group than expected in school A and more than expected in school B

E1

E1

2

[13]

Q20.

H_0 : The venue/location has no effect on the result of the match.

B1

O_i	E_i	$\frac{(O_i - E_i)^2}{E_i}$
15	10.8	1.6333
6	9.72	1.4237
6	6.48	0.0356
5	9.2	1.9174
12	8.28	1.6713
6	5.52	0.0417
$\Sigma O_i = 50$	$\Sigma E_i = 50$	$\chi^2_2 = 6.723$

calculation of E attempted

M1A1

Use of $\frac{(O_i - E_i)^2}{E_i}$

M1A1

AWFW 6.6 to 6.8

A1

$$\chi^2_{5\%}(2) = 5.991$$

For $v = 2$

B1

For 5.991 (on their v)

B1ft

$$5.991 < 6.72 \therefore \text{Reject } H_0$$

A1ft

Evidence suggest that the results are affected by the venue/location

E1ft

10

[10]

Q21.

H_0 : no association between their gender and type of holiday preferred
(for at least (H_0))

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H_1 : there is an association between their gender and type of holiday

Totals:

118	98	216
90	114	204
208	212	420

B1

Expected frequencies:

$$\frac{216 \times 208}{420} = 106.97$$

$$\frac{216 \times 212}{420} = 109.03$$

M1

$$\frac{204 \times 208}{420} = 101.03$$

A1

$$\frac{204 \times 212}{420} = 102.97$$

$$v = 1$$

B1

$$\sum \frac{[O_i - E_i - 0.5]^2}{E_i}$$

Yates' correction

M1

$$= 1.036 + 1.017 + 1.097 + 1.077$$

$$= 4.227$$

AWFW 4.20 to 4.24

A1

$$\chi^2_{5\%}(1) = 3.841$$

(4.20 if E_i rounded to nearest integer)

B1

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$$4.23 > 3.841 \therefore \text{reject } H_0$$

(Only if H_0 stated)

A1ft

Evidence at the 5% level of an association between the gender of single people and the type of holiday that they prefer to take.

E1ft

10

[10]

Q22.

H_0 : School & Examination grades are independent

B1

(i.e. no association)

$$v = (2 - 1)(4 - 1) = 3$$

B1

$$\chi^2_{5\%} = 7.815$$

on their v

B1f.t.

O_i	E_i	$\frac{(O_i - E_i)^2}{E_i}$
-------	-------	-----------------------------

52	49.8	0.0972
34	27.6	1.4841
16	23.4	2.3402
18	19.2	0.0750

Any correct method for E_i

All correct

M1

A1

114	116.2	0.0417
58	64.4	0.6360
62	54.6	1.0029
46	44.6	0.0321

400 400

Attempt at $\sum \frac{(O_i - E_i)^2}{E_i}$

M1

$$\chi^2 = 5.709$$

AWFW 5.70 to 5.71

A1

$$\left. \begin{array}{l} 5.709 < 7.815 \\ \therefore \text{do not reject } H_0 \end{array} \right\}$$

(Accept H_0)

M1f.t.

insufficient evidence at the 5% level of significance to suggest an association between school and examination grades

follow through on their values of X^2 and χ^2

A1f.t.

9

[9]

Q23.

H_0 : Gender of driver & speed of car are independent.

B1

H_1 : Gender of driver & speed of car are associated.

B1

$v = (3 - 1)(2 - 1)$
= 2 degrees of freedom

B1

O_i	E_i	$\frac{(O_i - E_i)^2}{E_i}$
17	29.3	5.5284(5.4980)
40	41.3	0.0394(0.0409)
70	55.9	3.5679(3.5565)

A1 for each correct row for E_i

M1A2

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30	17.2	9.6178(9.5256)
25	23.7	0.0685(0.0713)
18	32.1	6.2071(6.1935)

*attempt final column
allow 3sf answers*

M1

$\chi^2_{calc} = 25.04$ (24.89)

AWFW 24.8 to 25.1

A1

$\chi^2_{1\%}(2) = 9.210$

B1f.t.

$\therefore$ Reject H_0

Allow 'Accept H_1 '

B1f.t.

There is very strong evidence at the 1% level to suggest an association between the gender of the driver and the speed of the car

[11]

Q24.

H_0 : no association

OE

B1

120	100	220
24	36	60
144	136	280

correct totals

B1

calculate E_i :

$$\frac{220}{280} \times \frac{144}{280} \times 280 = 113.1$$

$$\frac{220}{280} \times \frac{136}{280} \times 280 = 106.9$$

$$\frac{220}{280} \times \frac{144}{280} \times 280 = 30.9$$

$$\frac{220}{280} \times \frac{144}{280} \times 280 = 29.1$$

Any correct method

M1

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

use of $\sum \frac{(O - E)^2}{E}$ (Ignore Y' C)

M1

6.357

40.4

0.3572
0.3782

attempted use of Yates' correction

M1

$$\begin{aligned}
 &1.3097 \\
 &1.3867 \\
 *Total &= 3.432
 \end{aligned}$$

$$\left. \begin{aligned}
 &AWFW(3.4 \text{ to } 3.5) \\
 &(NO Y' C \Rightarrow 3.993 \Rightarrow A0)
 \end{aligned} \right\}$$

(*) If wrong answer and no working out then M0 M0 A0

(*) If wrong answer and Yates Correction mentioned then M0 M1

A1

$$X_{5\%}^2(1) =$$

CAO ($v = 1$) $\Rightarrow$ by use of 3.841

B1

$$= 3.841$$

ft on their v

B1 f.t.

$\therefore$ do not reject H_0

These results do not suggest
an association between
driving schools and test results

Any correct interpretation dependent on 1st M1

A1 f.t.

EXAM PAPERS PRACTICE [9]

Q25.

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H_0 : Length of stay and frequency of mail are independent

H_1 : There is an association between the length of stay and frequency of mail
for Hypotheses

B1

$$\begin{aligned}
 v &= (3 - 1)(3 - 1) = 2 \times 2 = 4 \\
 &\text{i.e. 4 degrees of freedom}
 \end{aligned}$$

B1

	O_i	E_i	$O_i - E_i$	$\frac{(O_i - E_i)^2}{E_i}$
{	40	26.1	13.9	7.3418
	24	27.8	3.8	0.5078
	8	18.1	10.1	5.6364

{	12	16.34	4.3	1.1531
	18	17.35	0.7	0.0246
	15	11.31	3.7	1.2017
{	13	22.5	9.5	4.0204
	27	23.9	3.1	0.4022
	22	15.6	6.4	2.6389

$$\sum E_i = 179$$

A1 for each row (col) of E_i correct

M1A3

attempt final column

M1

$$X^2_{\text{CALC}} = 22.9$$

X^2_{CALC} (22.8 to 23.1)(rounded)

A1

$$X^2(4) = 13.277$$

(on their v)

B1f.t.

$\therefore$ Reject H_0 at 1% level
Accept H_1

The evidence suggests that there is a very strong association between the length of stay and the frequency of mail.

Any appropriate response

A1f.t.

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10

[10]

Q26.

(a)(b)

	Won	Lost	
< 2 hours	28 22.63	15 20.37	43
$\geq$ 2 hours	22 27.37	30 24.63	52
	50	45	95

Method for table

M1

Labelled correctly, allow >2

A1

Numerically correct

A1

Method for E's – their table

M1

3

H_0 No association between length of match and chance of Boris winning

H_1 Association between length of match and chance of Boris winning

Null hypothesis – may be implied by clearly stated conclusion – generous, allow 1-sided etc

B1

$$\sum \frac{(|O - E| - 0.5)^2}{E} = 4.04$$

Attempt at $\sum (O - E)^2 / E$

M1

Attempt at Yates' correction

m1

Correct application of Yates' Correction

m1

4.04 (4 ~ 4.10)

A1

c.v. $\chi^2_{1,2}$ is 3.841

1 df

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B1ft

3.841 or 3.84, their df

B1ft

Reject H_0 , evidence of association between result and length of game.

ft their figures – needs all M Marks and must be compared with upper tail of χ^2 (maximum 8 out of 9 if method of constructing table is incorrect) – Needs clearly stated correct conclusion or hypothesis

A1ft

9

- (c) There is evidence of association between result and length of game but contrary to

Boris's belief he appears to be less likely to win longer games.

evidence of association

E1

*Boris less likely to win long games (2 marks for this point if evidence of association clearly stated in (b))
Needs proportion probably implied for both marks*

E1

2

[14]

Q27.

(a)

	D		P		DN	
< 65	4	2.70	12	7.72	6	11.58
≥ 65	10	11.30	28	32.28	54	48.42

Method for E's

M1

Pooling D&P

	D&P		DN	
< 65	16	10.42	6	11.58
≥ 65	38	43.58	54	48.42

correct method for pooling - must be D&P

m1

H_0 : Answer not associated with age

B1

H_1 : Answer associated with age

correct null hypothesis - may be implied by clearly stated conclusion

$$\frac{(|O-E| - 0.5)^2}{E} =$$

attempt at $\sum \frac{(O-E)^2}{E}$ *- their figures*

M1

attempt at Yates' correction

m1

correct method for Yates'

m1

$$\frac{5.08^2}{\left(\frac{1}{10.42} + \frac{1}{11.58} + \frac{1}{43.58} + \frac{1}{48.42} \right)} = 5.83$$

5.83 (5.81 to 5.85)

A1

c.v. χ^2_1 is 3.841

1df or 2df if no pooling

B1

3.841 or 3.84 (allow 5.991/5.99 if no pooling)

B1

Reject H_0 , conclude evidence to suggest that answer is associated with age.

correct conclusion - must be compared with upper tail of χ^2

A1ft

10

- (b) Older delegates more likely to answer that they would not change their votes.

Older delegates less likely to answer that they would change their votes

E1

1

- (c) Most delegates are over 65

Most delegates are elderly

E1

1

[12]

Q28.

(a)

b	CC	IC	S
<120	2	16.94	58
≥ 120	34	19.06	12
		32.94	20
		37.06	44
			30.12
			33.88

Method for one E

M1 m1 B1

Method for all E's

M1 m1

H_0 : No association between asking price and area
 Correct H_0 - may be implied by conclusion

B1

H_1 : Asking price associated with area

$$\sum \frac{(O - E)^2}{E} = 67.3$$

Attempt at $\sum \frac{(O - E)^2}{E} = 67.3$

Completely correct method

67.3(67, 68) -allow if demonstrated to be greater than c.v.

M1 m1 A1

Critical value χ^2_{2df} for 1% risk 9.210
 ft 2df

B1ft

Reject H_0 , conclude asking price
 ft c.v. – candidate's df

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associated with area.

ft correct conclusion - must be compared with
 upper tail of χ^2
 (requires M1M1 only)

A1ft

9

- (b) Houses appear to be cheapest/proportion costing < £120 000 greatest, in inner city followed by suburban areas and most expensive in city centre
- Cheapest in inner city
 Most expensive in city centre

E1 E1

2

[11]

Q29.

(a)

	0	1	2	≥3	Total
<2	1478	212	21	6	1717
	1508.94	191.24	27		
			14.36	2.45	
			16.81		
>2	2830	334	21	1	3185
	2799.06	374.76	21		
			26.64	4.55	
			31.19		
Total	4308	546	41	7	4902

method for 4 E's

M1

Method for all E's

M1

Attempt to combine classes

M1

Correct method for combining classes - needs all previous M marks

m1

H_0 : No association between number of claims and time for which policy held.

H_1 : Number of claims associated with time for which policy held.

Correct H_0 - may be implied by correctly stated conclusion.

B1

$$\sum \frac{(O - E)^2}{E} = \frac{30.94^2}{1508.94} + \frac{30.94^2}{2799.06} +$$

Attempt at $\sum \frac{(O - E)^2}{E}$ - allow spurious continuity correction etc

M1

$$\frac{20.76^2}{191.24} + \frac{20.76^2}{354.76} + \frac{10.19^2}{16.81} + \frac{10.19^2}{31.19}$$

Correct use of $\sum \frac{(O - E)^2}{E}$ - requires first two and 4th M mark

m1

$$= 14.0$$

$$14.0 \text{ (13.9 to 14.1)}$$

A1

c.v. χ^2_2 for 1% risk is 9.21
2 df

B1f.t.

9.21, disallow ± 9.21

B1f.t.

Reject H_0 . Conclude number of claims associated with time policy held.

Conclusion - must be compared with upper tail of χ^2 .

A1f.t.

11

(b) (i), (ii)

A

Observed members not frequencies

E1

	Claim	No claim
Male	412	2824
Female	181	1485

cao

M1

A1

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Classes overlap

E1

	Claim	No claim
<25	280	676
≥ 25	314	3632

cao

M1

A1

C

Classes not complete

E1

9

	No claim	≤ £2000	>£2000
<25	676	135	145
≥ 25	3632	122	192

cao

M1

A1

[20]

Q30.

(a)

	12 – 3	3 – 6	6 – 9
12	22.58	6 12.51	54 36.92
62	51.42	35 28.49	67 84.08

Method for one E

Method for all E's

M1

M1

H₀: Alcohol independent of time of day

Correct null hypothesis - may be earned in conclusion

B1

H₁: Alcohol not independent of time of day

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$$\sum \frac{(O - E)^2}{E} = 23.4$$

Any attempt at $\sum \frac{(O - E)^2}{E}$

M1

Completely correct method

m1

23.4 (23.2 – 23.6)

A1

cv χ^2 for 1% 1 - sided risk is 9.210
2d.f.

B1

9.21 - upper 1% point of χ^2 , their d.f.

B1 ft.

Reject H_0 , conclude taking alcohol with meal associated with time of day

Conclusion, must use upper tail of χ^2 , depends on all method marks

A1 ft.

9

- (b) (i) A greater proportion of customers entering between 6 pm and 9 pm bring alcohol than those entering at other times
Clear interpretation gets both marks

E2(1)

- (ii) Customers enter at a greater rate between 6 pm and 9 pm than at other times
Customers enter at greater rate

E1

3

- (c) Ideally a random sample of all customers should be included but this is impractical. Conclusions only invalid if chosen day

E1

was not representative of all days e.g. if pattern was different at weekends.

Might affect conclusion

Reason

E1

2

[14]

Q31.

- (a) H_0 : Reason not associated with time of day
Correct null hypothesis – may be implied by clearly stated conclusion

B1

H_1 : Reason associated with time of day

	Rush hour		Out of RH	
Normal	34	27.05	45	51.95

Method for one E

M1

Beatles 4 10.95 28 21.05
Method for all E's

m1

$$\frac{\sum (O - E)^2}{E} =$$

$$\frac{\sum (O - E)^2}{E}$$

Attempt

Attempt Yates's correction

Clearly correct method for Yates's

M1 m1 m1

$$6.45^2 \left(\frac{1}{27.05} + \frac{1}{51.95} + \frac{1}{10.95} + \frac{1}{21.05} \right) = 8.11$$

8.11 (8.05 - 8.15)

A1

c.v. χ^2 is 3.841

1 d.f.

3.841 or 3.84

B1 B1ft

Reject H_0 conclude reason associated with time of day

Conclusion – t.s. must be compared with upper tail of χ^2 - needs first 3 M marks

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10

- (b) Of those using the crossing because of the Beatles, a greater proportion do so out of rush hour than during rush hour

*Greater **proportion** out of rush hour*

E1

1

[11]

Q32.

(a)

	J	R	S	
F	16 10.77	12 10.00	22 29.23	50
M	12 17.23	14 16.00	54 46.77	80

28 26 76 130

for at least one E
for all E's

M1 M1

H₀: Choice of language independent of gender
correct null hypothesis (may be implied by clearly stated conclusion)

B1

H₁: Choice of language not independent of gender

$\Sigma(O - E)^2/E = 5.23^2/10.77 + 5.23^2/17.23 + 2^2/10 + 2^2/16 + 7.23^2/29.23 + 7.23^2/46.77 = 7.68$
attempt at $\Sigma(O - E)^2/E$ using their E's - must be for all cells 7.6 ~ 7.7

M2 A1

c.v. χ^2 for 5% significance level is 5.991
2 degrees of freedom correct c.v., their degrees of freedom

B1 B1ft

Reject H₀, Conclude choice of language associated with gender.
correct conclusion their c.v. - test statistic must be compared with upper tail of χ^2 - needs all previous M marks.

A2ft

10

- (b) (i) Swedish more popular with males than with females

B1

1

- (ii) Swedish is most popular choice among both males and females
both marks for clearly expressed interpretation.

B2

2

[13]

Q33.

- (a) (i) H₀ : Irritation not associated with carrying phone
Correct null hypothesis - if not stated may be implied by final correct conclusion

H₁ : Irritation associated with carrying phone
Needs clear statement; can be earned in conclusion

or (a)(ii)

	Irritated		Not irritated	
Phone	12	17.03	14	8.97
No phone	26	20.97	6	11.03

Correct method for one E

B1

Correct method for all E's

M1 M1

$$\sum \frac{(|O - E| - 0.5)^2}{E}$$

Attempt $\sum \frac{(O - E)^2}{E}$

Attempt at Yates's correction

M1 m1

$$= 4.53^2 \left(\frac{1}{17.03} + \frac{1}{8.97} + \frac{1}{20.97} + \frac{1}{11.03} \right) = 6.33$$

Correct method for Yates's
(6.2 - 6.5) or demonstration

$$\sum \frac{(O - E)^2}{E} \text{ exceeds their c.v.}$$

m1 A1

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 cv $\chi^2_{1, 5\%} = 3.841$

for 1 degree of freedom
 3.841 or 3.84

B1 B1

Reject H_0 :irritation associated with carrying mobile phone.

Correct conclusion – requires first M1m1M1 and
 upper tail of χ^2 used

A1ft

10

- (ii) Passengers not carrying mobile phones more likely to be irritated than those carrying mobile phones

Conclusion in context

E1

- (b) (i) Combining dissimilar classes makes interpretation difficult or impossible

Interpretation difficult

E1

1

- (ii) Not necessary to combine extremely irritated as $E > 5$

Not necessary $E > 5$

E1

Anthony's suggestion leaves 3×2 table to be analysed which is more likely to establish an association if one exists (more degrees of freedom).

More likely to establish a difference / more degrees of freedom

E1

2

- (iii) H_0 : Irritation not associated with carrying phone

H_1 : Irritation associated with carrying phone

Null hypothesis - if not stated may be implied by final correct conclusion

B1

	Not / Slightly		Irritated		Extremely	
Phone	52	39.70	18	20.49	3	12.81
No Phone	10	22.30	14	11.51	17	7.19

$$\sum \frac{(O - E)^2}{E}$$

Attempt at $\sum \frac{(O - E)^2}{E}$

M1 M1

$$= \frac{12.30^2}{39.70} + \frac{12.30^2}{22.30} + \frac{2.49^2}{20.49} + \frac{2.49^2}{11.51} + \frac{9.81^2}{12.81} + \frac{9.81^2}{7.19} = 32.3$$

(32.2 - 32.5) or demonstration

$$\sum \frac{(O - E)^2}{E} \text{ exceeds their cv}$$

A1

CV $\chi^2_{21\%} = 9.210$

2 degrees of freedom

9.210 or 9.21

B1 B1

Reject H_0 , irritation associated with carrying phone.

Correct conclusion - requires previous M1 only and

upper tail of χ^2 used.

A1ft

Passengers not carrying mobile phones more irritated than others

Conclusion in context

E1

8

[22]



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Examiner reports

Q1.

In part (a), the vast majority of students scored full marks. Some students made a numerical error with one of the entries in the table but it was very rare to make more than one error. Some students truncated the value of 12.9375 to 12,937 as some calculators give the table with only five digits displayed for each number. Highlighting the numbers on the calculator and reading the full output on the calculator would avoid this error.

The majority of students scored at least four marks for part (b). The most common hypotheses stated by students were 'no association' for H_0 and 'association' for H_1 . Some students used 'independent' and 'not independent' instead, or a mixture. Some students failed to mention the variables in the hypotheses. Many students failed to merge the columns for 2 and 3+ even though the expected value of 4.375, which is less than 5, was given. Some students made numerical slips when calculating the test statistic. Many compared their test statistic with a critical value rather than calculating a corresponding probability. A minority of students drew a sketch to compare the test statistic with the critical value. Some sketches were not sufficiently clear and lost a mark.

Q5.

Most students followed the instruction to combine two columns, though some did not explain why this was necessary, and others ignored it completely. Only a small proportion thought it was appropriate to combine 'flats' with 'detached'; the vast majority making the obvious choice of 'flats' and 'terraced'. The calculations were generally well done, although some did not work with sufficient significant places. Only a few felt it necessary to use Yates' correction. The majority of students realised that the degrees of freedom had now been reduced to 2 and reached the correct conclusion. In the final part, when students chose difference between expected and observed values as their criterion, it was often unclear why 'terraced' might be chosen in preference to 'flats'. There were, however, many excellent explanations.

Q6.

Although the calculation of the test statistic was generally well done, with most students applying Yates' correction, very few truly appreciated what the χ^2 -test showed. The hypotheses were often in terms of the phone call, and even where the hypotheses were in terms of 'method of contact' the final conclusion was often "showing that a phone call is more effective". Only the best students realised that the χ^2 -test could only establish that an association between 'method of contact' and 'application for grant' existed, and that the values in the table must then be examined to support the council's belief. The majority correctly identified 'Type I error' and the reason for it.

Q7.

Part (a) was designed to test candidates' understanding of the properties of a contingency table. This proved to be very popular, with the majority of candidates showing a good understanding and demonstrating good algebraic skills to often achieve full marks. In part (b), most realised that Yates' correction was required even if they could not apply it correctly in all cases. However, this did, as usual, prove to be a popular source of marks, even for those candidates who found part (a) beyond them.

Q8.

As has been the case on previous papers, this was again one of the best answered questions. Most students managed to state correct hypotheses and then evaluate the required expected values. Most realised that 'degree classes 2(ii) and 3' needed to be combined and did so successfully, and for the correct reason that $E_i < 5$ for those students gaining a 'degree class 3' degree having gained an 'A-level grade A'. This usually led to the correct final column and hence the correct value of 13.5 for X^2 . Correct conclusions, in context, were then usually seen. Unfortunately some students, who realised that they had to combine in some way, decided incorrectly, to combine 'degree classes 1 and 3 or 2(i) and 3'. This obviously lost marks. Other students combined 'degree classes 2(i) and 2(ii)' and consequently lost most of the marks for the question whilst others combined for the wrong reason by indicating that $O_i < 5$ necessitated combining cells, or failed to combine any 'degree classes'. There were many excellent answers given to part (b). However, incorrect statements such as "Those receiving a class1 obtained fewer grade B's at A-level than expected" gained no credit.

Q9.

As in previous series, this proved to be a good source of marks for many candidates. In part (a), it was disappointing to see the many candidates who did not seem to understand the term 'contingency table'. Consequently, the 2 marks available were awarded to those candidates who filled in the table with the expected frequencies or retrospectively for those who calculated the expected frequencies correctly in part (b). Others, who simply copied the values from the table given in the question or who produced values which did not tally with the given totals, gained no credit.

In part (b), many excellent solutions were seen. However, in a few cases, the hypotheses were not stated and z or t values were stated instead of the required χ^2 value of 11.345. Also, rounding errors arising from the calculation of the expected frequencies often resulted in the calculated value of X^2 falling outside the acceptable range.

Q10.

As usual, the question on this topic was the best answered question on the paper with many fully correct solutions seen. The vast majority of candidates managed to correctly state at least the null hypothesis in part (a). In part (b), a few lost a mark for failing to give their answer to one decimal place, this despite the specific request. However, on the whole, this was very well done, often in conjunction with part (c) where the correct values of $v = 6$ and $\chi^2 = 16.812$ were usually seen, followed by a correct conclusion in context. Those candidates who stated an incorrect null hypothesis in part (a), or failed to find a value for X^2 within the range $11.9 \leq X^2 \leq 12.1$ in part (b), lost this mark. The responses to part (d) were better than usual but there was still room for improvement in this area. In part (e), many candidates found the correct value of $\chi^2 = 10.645$ but again lost the mark for the comment in context for the same reasons as in part (c).

Q11.

This was the second most common source of marks for the majority of candidates, with many fully-correct solutions seen. However, far too many candidates failed to state their hypotheses but still felt justified in drawing some sort of conclusion. The question required candidates to combine two rows. However, some candidates were thrown by the observed value of 2 which did not give an expected value of less than 5. This usually led to incorrect combining and even to the use of Yates' correction when a 2×2 table resulted. In fact, many candidates did not attempt to combine at all.

In part (b), only the more able candidates gave the correct interpretation that 'More

students than expected pass their test first time' or 'Fewer students than expected fail their test first time'. Many candidates appeared not to understand the question and so simply compared age groups.

Q12.

This was the most common source of marks for the majority of candidates, with many fully-correct solutions seen. However, too many candidates still failed to state their hypotheses but nevertheless felt justified in drawing some sort of conclusion. Since a 2×2 table was given, the use of Yates' correction was required, something that some candidates failed to do.

Q13.

This topic, as usual, proved to be the best source of marks on the paper, especially for those candidates who realised that when the *expected* frequency of any one category falls below 5 then this category *must* be combined with another. When combining, this should have been done with the category to which it was most closely related. In this case, combining Bulgarian with Polish to form a new category 'Languages' was the most sensible and, thankfully, the most often seen. Unfortunately, too many candidates simply combined the adjacent categories of Bulgarian and Climate Change or failed to combine at all.

A few candidates combined for the wrong reason by stating that "*observed* frequencies less than 5", or combined categories without having shown any evidence that expected frequencies had been calculated. The null hypothesis was usually stated correctly and the correct conclusion in context was often seen. However, some candidates were unable to state the correct critical value found from the χ^2 tables, whilst a few thought incorrectly that the use of a *t*-value or a *z*-value was appropriate. Some candidates, when quoting H_0 , seemed to think that the expressions 'not associated' and 'not independent' had the same meaning.

Q14.

This topic, as usual, proved to be the best source of marks on the paper, especially for those candidates who realised that when the expected frequency of any one category falls below 5 then this category must be combined with another. When combining, this should have been done with the category to which it was most closely related. In this case, the 16-17 and 18-21 age groups should have been combined, and also the 50-65 and Over 65 age groups, since each of these pairs of categories probably contained people with fairly similar and quite closely related views on the reorganisation of schools. It was not thought that combining the 16-17 age group with the Over 65 age group was sensible in the context of this question. Some candidates only combined once, either at the top end or at the bottom end of the age range and it was quite disappointing to see others not combining any categories at all with the consequential loss of marks.

The null hypothesis was usually stated correctly and the correct conclusion in context was often seen. However, some candidates were unable to state the correct critical value found from the χ^2 tables, whilst a few thought incorrectly that the use of a *t*-value or a *z*-value was appropriate.

Q15.

This, once again, proved to be a very popular topic and an excellent source of marks. Although there were still a few candidates who failed to state any hypotheses at all, those who did usually did so correctly. Most realised that $\nu = 1$ and that Yates' correction was

required. Unfortunately, there were some who could not apply this correction in the correct way. Although the majority of candidates correctly considered $|O_i - E_i| - 0.5 = 7.3$, and

then went on to calculate $\chi^2 = \sum_i \left(\frac{7.3^2}{E_i} \right)$, there were some who thought wrongly that $\chi^2 = \sum_i \left\{ \frac{(O_i - E_i - 0.5)^2}{E_i} \right\}$ or even that $\chi^2 = \sum_i \left\{ \frac{((O_i - E_i)^2 - 0.5)}{E_i} \right\}$ were appropriate.

It was pleasing to see good conclusions in context.

In part (b), most candidates understood what a Type I error was, stating that this occurred when “ H_0 was rejected when in fact H_0 was correct”. Unfortunately, many could not then interpret this Type I error in the context of the question.

Q16.

Although this first question provided most candidates with a good start to the paper, there were some who, in part (a), either failed to state any hypotheses or who thought that the null hypothesis should indicate ‘an association’, rather than ‘no association’, between the incidence of asthma and the volume of traffic. Also simply stating “ H_0 : not associated” and “ H_1 : associated” was not really sufficient when a statement in the context of the question was required. Although the great majority of candidates realised that Yates’ correction had to be used, there were far too many candidates who could not apply it correctly. A conclusion in context was required: simply stating “Reject H_0 ” was not sufficient to gain full marks. Although most candidates formulated a correct conclusion in context, some either used statements which were too positive in nature or made no statement at all. In part (b), the question asked for a comment about the 52 children who had asthma, given that they lived in an area where the volume of traffic was heavy. It did **not** ask, as many candidates thought, for a comparison with the children who lived in an area where the volume of traffic was light. Consequently, the required response “More than expected had asthma” was not always seen. Some candidates, having drawn the wrong conclusion in part (a) that there was ‘no association’ between the incidence of asthma and the volume of traffic, still felt justified in making this statement.

Q17.

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This type of question is always popular and this year it again proved to be a good source of marks. Most candidates correctly stated the null hypothesis and their conclusions were usually correctly stated in context. Many candidates used their calculator to perform all of

the necessary working and simply stated their value for $\sum_i \left(\frac{(O_i - E_i)^2}{E_i} \right)$ and gave the critical value of $\chi^2 = 10.645$ before then stating their conclusion in context. This is a method to be encouraged, especially for those candidates who can use their calculators proficiently. It should be understood, however, that the wrong value stated for

$\sum_i \left(\frac{(O_i - E_i)^2}{E_i} \right)$, together with no numerical calculations seen to indicate that a correct method has been used, will usually result in a loss of many marks. The responses to part (b) were rather mixed, with the correct answer “More of those students achieving level 7 at KS3 gained a grade A at GCE than was expected” often being seen. However, there were many general comments seen that did not relate to this group of students at all.

Q18.

This question, as usual on this topic, proved to be a good source of marks for the majority of candidates. However, there were still some candidates who either did not state their hypotheses in context or omitted them completely. Simply stating “H₀: not associated” and “H₁: associated” was not sufficient. Although the vast majority of candidates realised that Yates’s correction had to be used, there was a great number of candidates who could not apply it correctly. The most common incorrect applications were $O_i - E_i - 0.5$ or $(O_i - E_i)^2 - 0.5$, whereas $|O_i - E_i| - 0.5$ was required. Although most candidates formulated a conclusion in context, these were usually too positive in nature. Also, simply stating “Reject H₀” was not sufficient to gain full marks.

Q19.

Many candidates failed to realise that frequencies and not percentages must be used when constructing a contingency table. Consequently, in part (a)(i), these candidates gained no credit. The majority of candidates were able to demonstrate the required techniques in part (a)(ii), albeit often using the given percentage values. Sensible conclusions in context and related to stated hypotheses were usually seen. In part (b), most candidates failed to answer the question set. Candidates were expected to compare observed and expected values. Instead, most simply made general comments about the 11-24 age group and, as a consequence, gained no credit.

Q20.

This was the best answered question with full marks gained by about half of the candidates. It should be noted that a statement of at least the null hypothesis is expected in questions of this type, with conclusions in context.

Q21.

Solutions to this question were, as usual, mostly excellent. However, some candidates still failed to state their hypotheses whilst others got H₀ and H₁ the wrong way round. Candidates should realise that stating ‘reject H₀’ without first stating H₀ is inappropriate. Also expected values should not be rounded to whole numbers especially when the value of χ^2 is then quoted to two or more places of decimals. Few candidates failed to attempt to

use Yates’ correction but several candidates considered $\frac{(O_i - E_i - 0.5)^2}{E_i}$ rather than the

correct form of $\frac{(|O_i - E_i| - 0.5)^2}{E_i}$ when applying the correction. A statement of a conclusion in context was expected but was not always seen.

Q22.

Many excellent and fully correct solutions were seen to what has proved to be a very popular topic, even with weaker candidates. The main errors usually came in the initial stating of the null hypothesis and statement of the final conclusions which were again either too positive or not given in context. The calculations were often completely correct with only a few mistakes seen in the determination of the number of degrees of freedom or in the use of the χ^2 tables.

Q23.

This proved to be the most popular question on the paper and many excellent solutions were seen. Hypotheses were usually stated correctly and conclusions stated in context.

Candidates did tend to be far too positive when forming their conclusions, not realising that the rejection of H_0 does not mean that H_1 is true. Conclusions should be stated in the form 'There is evidence, at the 1% level of significance, to suggest an association ...'.

Some candidates failed to quote the value of $\chi^2_{v^2}(1\%)$ and consequently lost marks.

Q24.

Candidates scored consistently well on this question, with many gaining full marks. However, some still failed to state their hypotheses while others wrongly assumed that H_0 indicated an association between test results and the school chosen.

Only a few candidates omitted the use of Yates' correction whilst even fewer did not state $v = 1$, leading to a χ^2 value of 3.841.

Candidates should be encouraged to write their conclusions in the context of the problem set. Statements such as 'do not reject H_0 ' or 'accept H_1 ', without reference to the context of the question are not likely to gain full credit.

Q25.

This question was generally well answered with many candidates presenting fully correct solutions and even the weakest candidates gaining some marks. Unfortunately, there were some candidates who did not write down their hypotheses even though the question specifically requested them to do so. Some candidates attempted to use Yates' correction even though this was inappropriate. Most candidates were able to correctly state $v = 4$ but often gave the 10% value rather than the required 1% value of 13.277. Some candidates truncated the value given in the table and consequently lost credit. Final conclusions had to be made in context, although here "reject H_0 " was accepted.

Q26.

As usual this type of question was extremely well answered with the great majority of candidates constructing the contingency table correctly and most applying Yates' correction accurately. The most common error with Yates' correction was to use $(O - E - 0.5)$ instead of $(|O - E| - 0.5)$. There were some problems with the null hypothesis. Apart from some nonsensical hypotheses such as 'no difference between the length of the game and the probability of winning' there were also some one-sided hypotheses such as 'the longer a match lasts the better the chance of Boris winning'. The hypothesis test can only test for 'association' although further interpretation is possible by examining the table.

Q27.

As usual candidates were well able to deal with contingency tables although some lost marks by failing to pool. A few pooled without first calculating any expected values, implying that they had based their decision to pool on the observed rather than the expected values.

Q28.

The great majority of candidates answered part (a) successfully. A few incorrectly thought that the small observed value for city centre properties costing less than £120 000 made it necessary to combine classes. In part (b), many merely repeated their conclusion to part (a) instead of observing that properties appear to be most expensive in the city centre and cheapest in the inner city.

Q29.

Candidates performed very well on this question. Part (b) was testing, but most candidates were able to make a reasonable attempt.

Q30.

The performance of the candidates, as a whole, in analysing the contingency table can only be described as excellent. To gain full marks in parts (b)(i) and (b)(ii) it was necessary to distinguish clearly between the *proportion* and the *number* of customers bringing alcoholic drinks.

Q31.

Nearly all candidates were able to make a good attempt at analysing the contingency table. Only a few failed to attempt Yates' correction, although the use of

$\sum \frac{(O - E - 0.5)^2}{E}$ instead of $\sum \frac{(|O - E| - 0.5)^2}{E}$ was not uncommon. However, the interpretation in part (b) was not well done. The expected answer was that the proportion of people using the crossing because of the Beatles was less in rush hour than outside rush hour.

Q32.

The great majority of candidates scored high marks by carrying out the calculations correctly although some then drew the wrong conclusion. Most offered some relevant interpretation but only a minority gained full marks for part (b). A comment made by many candidates was that more males than females studied Swedish. This is true but, as there were more male than female students, it was hardly surprising. The data and the test result implied that a significantly greater proportion of males than females studied Swedish.

Q33.

Most candidates scored highly on this question. The mechanics of analysing contingency tables, including, where appropriate, combining classes and the use of Yates's correction, seems well understood. Many had difficulty framing appropriate hypotheses. 'H₀ not linked: H₁ linked' is not adequate, although candidates were allowed to redeem themselves by a clearly stated conclusion which enlarged on exactly what was linked. In part (a)(ii), candidates were expected not only to state that irritation and carrying a mobile phone were associated but that (unsurprisingly) passengers not carrying a mobile phone were more likely to be irritated than those carrying a mobile phone.

In parts (b)(i) and (b)(ii), many candidates scored marks by having more or less the right ideas despite some difficulty in expressing them convincingly. In part (b)(i), most candidates thought it was a bad idea to combine opposites while not going on to comment on the difficulty this would cause when trying to interpret the results. In part (b)(ii), most candidates thought a 3 × 2 table was preferable to a 2 × 2 table, without really being sure why. Vague reasons such as 'more accurate' were given. More substantial reasons such as 'less likely to lead to a Type 2 error' were much rarer.